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Underwriting: Starter for 10 (minutes !)

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Underwriting explained in 10 minutes? It's going to be....



What does an underwriter do ?

Keep claims assessors in business



What does an Underwriter actually do?

Work closely with actuaries, risk and claims managers to ensure a balance between attracting and retaining customers through competitive insurance premiums (fees) and being able to cover any potential losses from claims

'Insurance Underwriter' A financial professional that evaluates the risks of insuring a particular person or asset and uses that information to set premium pricing for **insurance** policies

Assess risk

Assess a risk according to the likelihood of a claim being made by weighing up a number of factors and asking for detailed information from prospective clients (policyholders)

Decide if applications for insurance cover (risks) should be accepted and, if so, what the terms of that acceptance are

Why Underwrite?

The reasons for underwriting are to:

Maintain premium rates at a competitive level

Offer a fair premium to all policyholders

Ensure the company does not place itself at excessive risk

Minimise the risk of fraudulent claims

Minimise the risk of adverse claims experience arising from anti selection

Minimise the risk of early lapses

Underwriting - Isn't it all done by computer nowadays?

Yes, most of it is done via an Underwriting rules engine. Typically, around 75% of a company's business may be underwritten this way.



Very clever & sophisticated design – you put the application in, turn the handle and out pops a decision



At a very high level, rules engines have 'straight through processing' (STP) rates ranging from 50-75% (system provided a decision: standard rates, rated, exclusion, postpone/decline)

However, we are going to take a closer look at underwriting, why we do it and I'm going to build a 'visual risk profile' ! Meet 'Bob' !

Meet 'Bob' Blyth.....

Bob's a 36 year old male, who's moving house and wants insurance to cover his mortgage > £450,000 Life and CI cover over 15 years



On the face of it, he looks a standard rates risk. However, let's see what happens when we look at some (not all) of his application form questions and answers..... building that 'visual risk profile ' along the way!

This will give you a 'brief flavour' of why the underwriting process asks some of the questions it does.

Occupation... What are the risks?

Typical application form question:

Does your occupation involve working in any of the following:-
Armed forces (including reservists/territorial army),
Aviation with flying duties, Fishing, Oil/gas industry (offshore),
Underwater, With explosives, Heights over 12 metres?

☐

Yes

☐

No

☐

Yes

If you have answered 'Yes' to the above question, please provide full details in the box provided below.

Why could this be a concern for underwriting?

- ☐ Bobs occupation is a builder, working at heights over 12 metres (40 feet) at least 60% of his time



Risks associated with construction industry:

- Working at heights (usually 1/3 deaths)
- Slips/falls/trips
- Moving objects
- Collapse (excavations/trenches etc.)
- Asbestos
- Airborne Fibres & Materials= Respiratory disease (COPD etc.)
- Electricity

Pastimes... What are the risks? (Diving)

Typical application form question:

Do you, or do you intend to take part in any hazardous sports or activities such as mountaineering, diving, private flying, motor sport etc?

Why could this be a concern for underwriting?

- ❑ Bob is a qualified Scuba diver, approx. 10-15 dives pa, Max 30m, no wreck /cave/ice diving



Risks associated with diving:

- Malfunctioning Equipment
- Pulmonary Embolism (rapid ascent)
- Nitrogen Narcosis
- Marine Life
- Lack of fitness
- Wreck diving (entering the wrecks)
- Cave Diving
- Ice Diving
- Increased depth
- Drowning
- Poor visibility
- Entanglement
- Temperature (ie Hypothermia)

Lifestyle... What are the risks? (Obesity)

Typical application form question:

Height and weight

What is your height without shoes?

ft

ins

or

cms

What is your weight in indoor clothes?

st

lbs

or

kgs

Why could this be a concern for underwriting?

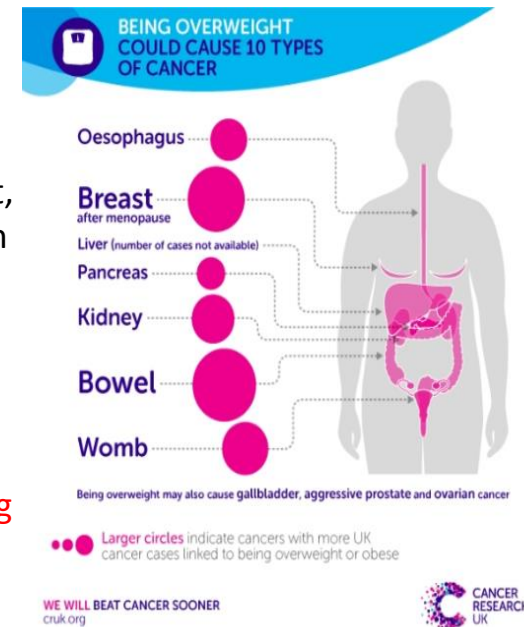
❑ Bob is 5ft 6 inches (168cms) and weighs 93kgs (205lbs) BMI = 33



Risks associated with obesity:

Obesity is a health concern. Without proper treatment, obesity can lead to other serious health problems, such as:

- osteoarthritis
- heart disease and blood lipid abnormalities
- stroke
- type 2 diabetes
- sleep apnoea (when you periodically stop breathing during sleep)
- reproductive problems
- gallstones
- **certain cancers**
- obesity hypoventilation syndrome
- metabolic syndrome



Read more at <http://www.cancerresearchuk.org/about-cancer/causes-of-cancer/bodyweight-and-cancer/how-being-overweight-causes-cancer>

Lifestyle... What are the risks? (Smoking)

Typical application form question:

Have you smoked or used any tobacco or nicotine products in the last 12 months?

☐ Yes ☐ No ☐ Yes ☐ No

Note: If you answer 'No' to this question, you may be asked to undergo a test to verify your answer.

For the following tobacco products, please state your typical consumption a day.

	Cigarettes		Cigarettes				
	Cigars		Cigars				
Pipe tobacco							
	ounces	or		ounces	or		grams
	grams			grams			grams

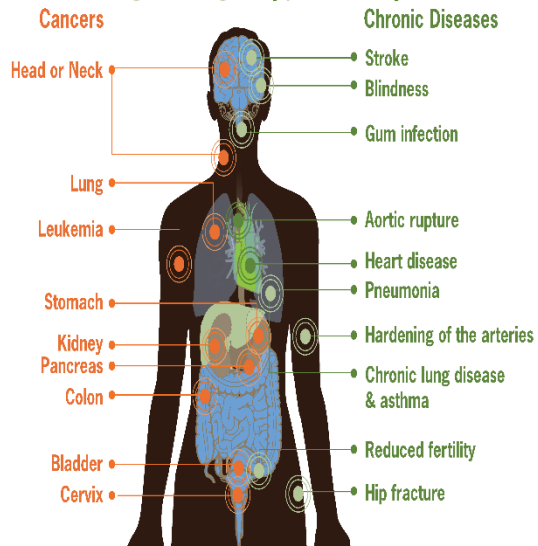
Why could this be a concern for underwriting?

☐ Bob is a 'social smoker' and smokes 5-10 cpd (he is very sociable!)



Risks from Smoking

Smoking can damage every part of the body



Risks associated with smoking:

- Smoking causes about 90% of **lung cancers**. It also causes cancer in many other parts of the body, including the: **mouth, lips, throat, voice box (larynx) oesophagus (the tube between your mouth and stomach), bladder, kidney, liver, stomach, pancreas**
- Smoking damages your heart and your blood circulation, increasing your risk of developing conditions such as: **coronary heart disease, heart attack, stroke, peripheral vascular disease (damaged blood vessels), cerebrovascular disease (damaged arteries that supply blood to your brain)**
- Smoking also damages your lungs, leading to conditions such as: **chronic obstructive pulmonary disease (COPD), which incorporates bronchitis and emphysema, pneumonia**

Source: NHS choices.

Lifestyle... What are the risks? (Recreational drugs)

Typical application form question:

Have you ever used recreational drugs
(e.g. cannabis, cocaine, heroin)

☐ Yes

☐ No

Why could this be a concern for underwriting?

☐ Bob likes to occasionally 'relax' by smoking some weed ! (not much - couple of times a week)



Street names for cannabis:

Weed, Pot, Grass, Ganja, Skun,

Hashish/Charas , Bhang,

Hash oil, Reefer, Herb, Boom, Blubbers, Gangster

Risks associated with cannabis use:

- Using it has also been linked, in some people, to serious, long-term mental health problems
- Tobacco and cannabis share some of the same chemical 'nasties' and just like smoking tobacco, smoking cannabis has been linked to lung diseases like tuberculosis and lung cancer
- Mixing cannabis with alcohol can have particularly serious consequences - the accident rate is 16 times higher than for cannabis or alcohol alone
- It can increase the heart rate and affect blood pressure, which can be especially harmful for those with heart disease
- A recent review of cannabis research published in the British Medical Journal found those driving under the influence of cannabis had nearly double the risk of a crash

Lifestyle... What are the risks? (Alcohol)

Typical application form question:

Alcohol consumption

How many units of alcohol do you drink on average each week?
(One pint of beer = 2.5 units, one 330ml bottle of beer = 1.5 units,
one 175ml glass of wine = 2 units, one measure of spirits = 1 unit).
If you don't drink alcohol please enter 0.

units

Why could this be a concern for underwriting?

❑ Bob likes to occasionally 'relax' by having a drink or two a few times a week (at least 30 units)



*The government's alcohol guidelines changed in Jan 2016 to reflect new evidence about the link between alcohol and health harms, particularly cancer.

Risks associated with alcohol use:

Your liver processes alcohol. It can only cope with so much at a time. Drinking more alcohol than the liver can cope with can damage liver cells and produce toxic by-product chemicals. The more you drink, and especially above the recommended limits, the greater the risk of developing or experiencing serious problems such as:

- Liver disease (cirrhosis or hepatitis)
- *Cancer (recent study linked to 7 cancers -oropharynx, larynx, oesophagus, liver, colon, rectum and breast)
- Gut and pancreas disorders
- Depression / Anxiety
- Muscle and heart muscle disease
- High blood pressure
- Damage to nervous tissue
- Serious accidents
- Obesity (alcohol is calorie-rich)



Medical questions... What are the risks? (High Blood Pressure)

Typical application form question: Apart from anything you have already told us about, DURING THE LAST 5 YEARS have you seen a doctor, nurse or other health professional for:

- raised blood pressure, raised cholesterol, or condition affecting blood or blood vessels, for example anaemia, excess sugar in the blood, blood clot, deep vein thrombosis?

Why could this be a concern for underwriting?

- ☐ Bob answered all application medical questions 'no' expect for advising he is on treatment for raised blood pressure. He is on Atenolol and his GP advised his last reading was 'normal' 3 months ago.



Risks associated with High Blood Pressure:

High blood pressure is not usually something that you can feel or notice, but over time if it is not treated, your heart may become enlarged making your heart pump less effectively. This can lead to **heart failure**.

Having high blood pressure increases your chance of having a heart attack or stroke. There isn't always an explanation for the cause of high blood pressure, but these can play a part:

- not doing enough physical activity
- being overweight or obese
- having too much salt in your diet
- regularly drinking too much alcohol or
- having a family history of high blood pressure.

The 'visual risk profile'..... Bob & his 'risks'

Does he still look a standard rates risk?



The 'visual risk profile'....Bob & his 'risks' – The UW manual

What would we do ? Probably ask a host of questions and get more medical evidence, an examination and perhaps Liver function tests! - however for the purpose of this session, we looked at the **UW manual**:

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SOLEM



*CI-exclude
disability benefits

Appreciation elements			Life	Critical Illness
Carpenter		☐	Standard	Standard
Ceiling fixer		☐	Standard	Standard
Fence erector		☐	Standard	Standard
Timber structure assembler		☐	Standard	Standard
Structural steel worker		☐	+2 PM	Exclusion (1)
Structural steel erector		☐	+2 PM	Exclusion (1)
Construction machinery operator				
Jackhammer operator		☐	Standard	Exclusion (1)
Hammerman		☐	Standard	Exclusion (1)
Hoist / winch operator		☐	Standard	Standard
Air compressor operator		☐	Standard	Standard
Pile driver operator		☐	Standard	Exclusion (1)
Crane operator - Sectors excluding oil-gas		☐	+1 PM	Exclusion (1)
Demolition of buildings				
Without use of explosives		☐	+1 PM	Exclusion (1)
With use of explosives		☐	+2.50 PM	Exclusion (1)
Bricklayer		☐	Standard	Standard
Bricklayer's mate / Hod carrier		☐	Standard	Exclusion (1)
Concreter				
Erector		☐	+2 PM	Exclusion (1)
Finisher		☐	Standard	Standard
Earthwork contractor				
Without manual work		☐	Standard	Standard
With manual work		☐	Standard	Exclusion (1)

The 'visual risk profile'.... Bob & his 'risks' – The UW manual



SCUBA Diving		
Practised alone and without qualified supervision	Decline	Decline
Diving without a medical fitness certificate (issued annually)	Decline	Decline
Practised in a group		
Without certification		
With an instructor	Standard	Standard
Without an instructor	Exclusion (1)	Exclusion (1)
With certification		
Depth <40 m (1)		
< 5 dives a year	Standard	Standard
5 - 19 dives a year	0.50‰	0.50‰
≥ 20	0.50‰	0.50‰
Depth 40 - 60 m (usually requires Nitrox or other gas mixtures)	2.00‰	Decline
Depth 61 - 80 m (Nitrox or other gas mixtures)	4.00‰	Decline
Depth > 80 m (1)	Decline	Decline



C.V. Risks

Info

Description

General Information

Age years
 Sex ☒ Male ☐ Female
 Tariff ☒ Aggregate ☐ Smoker / non smoker

Body Mass Index

Unit ☒ Metric ☐ Imperial
 Height cm
 Weight Kg
 BMI

	Life	Critical Illness
Final Result	+25	+50

Tobacco use

Smoker ☒ yes ☐ no ☐ unknown
 Cigarettes/day

	Life	Critical Illness	TPD Own
Final Result	+0	+25	+0

Final Result

	Life	Critical Illness	TPD Own
Final Result	+25	+75	+75

The 'visual risk profile'.... Bob & his 'risks' – The UW manual



Present or < 1 year after last use			
With sequelae, e.g. nasopharyngeal or bronchopulmonary disorders	☐	Decline	Decline
With pre-existing neurotic or psychotic disorders or unstable socio-economic situation	☐	Decline	Decline
No pre-existing neurotic or psychotic disorders, stable socio-economic situation and no sequelae			
Not combined with the use of other drugs or substances			
Less than one use per week	☐	Standard	Postpone
1 - 3 uses per week	☐	+50	Postpone
More than 3 uses per week	☐	Decline	Decline
Combined with other drugs or substances	☐	Decline	Decline



Diagnosis of alcoholism, confirmed abstinence: Rate as <u>Alcoholism</u>			
Associated thrombocytopenia (platelet count < 140,000/mm ³ or <140 10 ⁹ /L	☐	Postpone	Postpone
Criticism of consumption - never diagnosed as an alcoholic			
Mild - consumption less than 35 units per week, no symptoms or liver enlargement and LFTs < 2x normal ?	☐	+50	+75
Moderate - consumption between 36 and 56 units per week, or occasional associated symptoms (eg dyspepsia, gastritis), or liver enlargement up to 2 fingers breadth, or LFTs 2-4x normal ?	☐	+150	Decline
Severe - consumption greater than 56 units per week, or persistent associated symptoms (eg dyspepsia, gastritis), or liver enlargement greater than 2 fingers breadth, or LFTs more than 4x normal ?	☐	Decline	Decline



Blood Pressure

	Systolic BP	Diastolic BP
1st reading	130	90 mmHg
2nd reading		mmHg
3rd reading		mmHg
		Life
		Critical Illness
Final Result	Standard	Standard

The 'visual risk profile'..... Bob & his 'risks'

What would I do?

LIFE COVER

CI COVER



Accept at standard rates ?



Rate and apply extra premium or ask further questions ?



Postpone or decline ?



What's Bob main risk?

Coronary Heart Disease:

Every 3 minutes, someone in the UK is struck by a heart attack. Around 30% of heart attacks are fatal. Around 2.3 million people in the UK are living with coronary heart disease.. *Source British Heart Foundation.*

Coronary heart disease (CHD) is when the coronary arteries (the arteries that supply the heart muscle with oxygen-rich blood) become narrowed by a gradual build-up of fatty material within their walls. This condition is called **atherosclerosis** and the fatty material is called atheroma. In time, the arteries may become so narrow that they cannot deliver enough oxygen-rich blood to the heart. The pain and discomfort felt as a result is called angina.

If a piece of atheroma breaks off it may cause a blood clot (blockage) to form. If it blocks the coronary artery and cuts off the supply of oxygen-rich blood to the heart muscle, the heart may become permanently damaged. This is known as a heart attack.

What increases the risk of CHD?

There are several risk factors for CHD. The main ones are:

- Smoking
- High blood pressure
- High blood cholesterol
- Diabetes
- Being physically inactive
- Being overweight or obese
- Family history of heart disease
- Ethnic background
- Sex - men are more likely to develop CHD at an earlier age than women
- Age - the older, the more likely to develop CHD.

The more risk factors someone has , the more likely they are to develop CHD



Thanks for listening
and
hopefully keeping up !